

Name: Romulo Reyes Jr
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Course: Foundation Of Python Programming

Assignment05: Student Course Registration with Error Handling and Input Validation Program

Introduction

This Student Course Registration with error handling and input validation Program is a simple console-based Python application that lets users register students for courses, view current enrollments, and save the enrollment list to a JSON file (Enrollments.json) stored alongside the script. It validates first and last names to contain only letters, keeps enrollments in memory as a list of dictionaries (FirstName, LastName, Course), and provides a clear menu-driven workflow with options to add entries, display current data, persist data to disk, and exit. The program is designed for small-scale use, quick testing, and as a starting point for adding features like duplicate checks, improved validation, or automatic save-on-exit.

Student Course Registration with Error Handling and Input Validation Program

The Student Course Registration with an error handling and input validation program is to facilitate the enrollment of students in various courses. It allows users to register students by collecting their names and chosen course details, display the current list of enrolled students, and save this enrollment data to a JSON file for persistent storage. The program ensures data validation and manages file operations effectively.

The main features of the Course Registration Program include:

1. **Student Registration:** Users can register students by entering their first name, last name, and course name, which are then stored in a structured format.

```
---- Course Registration Program ----
Select from the following menu:
  1. Register a Student for a Course
  2. Show current data
  3. Save data to a file
  4. Exit the program
-----

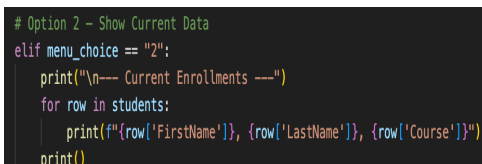
Enter your choice [1-4]: 1
Enter student's first name: Romulo
Enter student's last name: Reyes
Enter course name: Python 100
Student registered successfully.
```

2. **Data Display:** The program can display the current list of enrolled students, showing their names and the courses they are registered for.

```
---- Course Registration Program ----
Select from the following menu:
  1. Register a Student for a Course
  2. Show current data
  3. Save data to a file
  4. Exit the program
-----

Enter your choice [1-4]: 2

--- Current Enrollments ---
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```



```
# Option 2 - Show Current Data
elif menu_choice == "2":
    print("\n--- Current Enrollments ---")
    for row in students:
        print(f"{row['FirstName']}, {row['LastName']}, {row['Course']}")
    print()
```

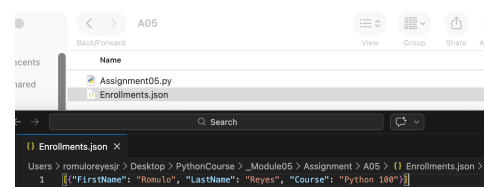
3. **Data Persistence:** Enrollment data can be saved to a JSON file (Enrollments.json), allowing for data retention across program executions.

```
----- Course Registration Program -----
Select from the following menu:
1. Register a Student for a Course
2. Show current data
3. Save data to a file
4. Exit the program
```

Enter your choice [1-4]: 3

Data successfully saved to file.
Saved Data:
Romulo, Reyes, Python 100

```
# Option 3 - Save Data to File
elif menu_choice == "3":
    try:
        file = open(FILE_NAME, "w")
        json.dump(students, file)
        file.close()
        print("\nData successfully saved to file.")
        print("Saved Data:")
        for row in students:
            print(f"{row['FirstName']}, {row['LastName']}, {row['Course']}")
        print()
    except TypeError:
        print("Type error occurred while writing to file.")
    except Exception as e:
        print(f"Unexpected error occurred while writing file: {e}")
```



4. **Input Validation:** The program validates user inputs to ensure that names contain only alphabetic characters, enhancing data integrity.

Enter your choice [1-4]: 1
Enter student's first name: Cheryl
Enter student's last name: Reyes582

Error: Last name must contain only letters.

```
----- Course Registration Program -----
Select from the following menu:
1. Register a Student for a Course
2. Show current data
3. Save data to a file
4. Exit the program
```

Enter your choice [1-4]:

```
# Option 1 - Register Student
if menu_choice == "1":
    try:
        student_first_name = input("Enter student's first name: ").strip()
        if not all(c.isalpha() or c.isspace() for c in student_first_name):
            raise ValueError("First name must contain only letters.")
    except ValueError as e:
        print(f"\n{e}")
        continue

    try:
        student_last_name = input("Enter student's last name: ").strip()
        if not all(c.isalpha() or c.isspace() for c in student_last_name):
            raise ValueError("Last name must contain only letters.")
    except ValueError as e:
        print(f"\n{e}")
        continue

    course_name = input("Enter course name: ").strip()

    student_data = {
        "firstName": student_first_name,
        "lastName": student_last_name,
        "Course": course_name
    }

    students.append(student_data)
    print("Student registered successfully.")
```

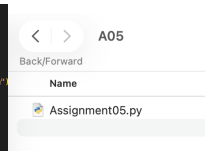
5. **Error Handling:** It includes error handling for file operations and input validation, providing user-friendly messages in case of issues (e.g., file not found, invalid JSON).

Warning:Enrollments.json not found. Starting with empty data!

```
----- Course Registration Program -----
Select from the following menu:
1. Register a Student for a Course
2. Show current data
3. Save data to a file
4. Exit the program
```

Enter your choice [1-4]:

```
# Load existing data from file at program start
try:
    file = open(FILE_NAME, "r")
    students = json.load(file)
except FileNotFoundError:
    print(f"\n{e}")
except json.JSONDecodeError:
    print(f"\n{e}")
except Exception as e:
    print(f"\n{e}")
    file.close()
```



6. **User-Friendly Interface:** The program features a simple command-line menu that guides users through various options, making it easy to navigate.

```
----- Course Registration Program -----
Select from the following menu:
1. Register a Student for a Course
2. Show current data
3. Save data to a file
4. Exit the program
```

```
# MENU str = """
----- Course Registration Program -----
Select from the following menu:
1. Register a Student for a Course
2. Show current data
3. Save data to a file
4. Exit the program
"""
```

7. **Graceful Exit:** Users can exit the program smoothly while receiving a confirmation message.

```
----- Course Registration Program -----
Select from the following menu:
1. Register a Student for a Course
2. Show current data
3. Save data to a file
4. Exit the program
```

Enter your choice [1-4]: 4
Program exiting...Goodbye!

```
# Option 4 - Exit
elif menu_choice == "4":
    print("Program exiting...Goodbye!")
    break

else:
    print("Invalid option, please choose between 1-4.\n")
```

These features work together to create an efficient and user-friendly course registration system.

Summary

This Course Registration with error handling and input validation Program is a simple yet effective tool for managing student enrollments by leveraging JSON for data storage and the program implements error handling for file operations (FileNotFoundError, JSONDecodeError) and input validation (ValueError) to ensure robustness and user-friendly messages providing a basic command-line interface for user interactions.